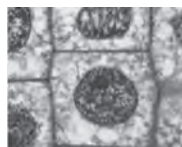


2 Interphase and mitosis are parts of the cell cycle.

(a) The photograph shows a plant cell in interphase.



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(i) Interphase is the part of the cell cycle when mitosis is not taking place.

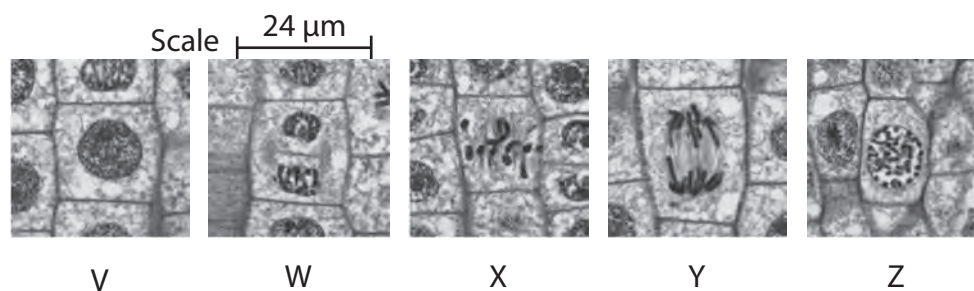
Which of the following processes occur during interphase?

1. cell growth
2. genetic material is replicated
3. organelles are synthesised

(1)

- ☐ **A** 1 only
- ☐ **B** 1 and 2 only
- ☐ **C** 2 and 3 only
- ☐ **D** 1, 2 and 3

(ii) The photographs show some stages in the cell cycle.



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Which row shows the correct sequence of these stages in the cell cycle?

(1)

Sequence of stages					
☐ A	V	X	W	Y	Z
☐ B	V	X	Y	Z	W
☐ C	V	Z	X	Y	W
☐ D	V	Z	W	X	Y



(iii) Calculate the magnification of photograph **W**.

Use the scale bar to help you.

(1)

Answer

(b) The mitotic index of a tissue was calculated.

The tissue had 450 cells in mitosis and the mitotic index was 37.5%.

Calculate the number of cells in this tissue that were in **interphase**.

(2)

Answer

(c) Describe the events that occur during prophase of mitosis.

(2)

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(d) Some new cells produced by mitosis are used for repairing damaged tissues.

Some new cells produced by mitosis are used for growth of tissues.

(i) Give an example of a tissue that could be produced in a plant.

(1)

(ii) Give **one role**, other than for growth or repair, for new cells produced by mitosis.

(1)

(Total for Question 2 = 9 marks)

